Task 8 - Trace the Compilation Process

C Programming Assignment - Rural Electricity Billing System

Four stages

Preprocessing expands includes and macros; compilation translates C to assembly; assembly creates object files; linking combines object files and libraries into an executable.

Commands

gcc -E main.c -o main.i; gcc -S main.c -o main.s; gcc -c main.c -o main.o; gcc main.o billing.o -o billing.

Single vs multiple files

One file is simpler for a small program. Multiple files separate interfaces, implementation and input/output. Each source file is compiled separately and then linked.

Warnings and debugging

Use gcc -std=c11 -Wall -Wextra -Wpedantic and fix warnings. Use -g for debugging and consider -O2 for an optimized build after testing.

Complexity

Compile time depends on source size, included headers and optimization. Runtime complexity is a separate measurement.

Important: illustrative tariff rates are used because the screenshot does not show the complete official tariff table. Replace them with your instructor's prescribed rates.